

Théo Matricon

2025–current

Postdoc with Mathieu Acher

Code Generation, Reproducibility

DiverSE, Software Engineering team

IRISA, Rennes · **medical condition** (RQTH)

2021–2024

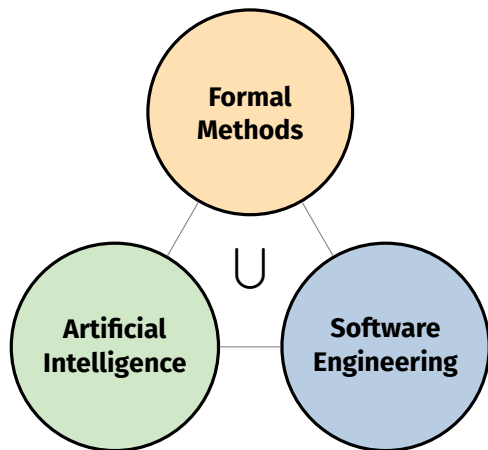
PhD supervised by Nathanaël Fijalkow

Scaling domain agnostic techniques for program synthesis

M2F, Formal Methods team

LaBRI, Bordeaux

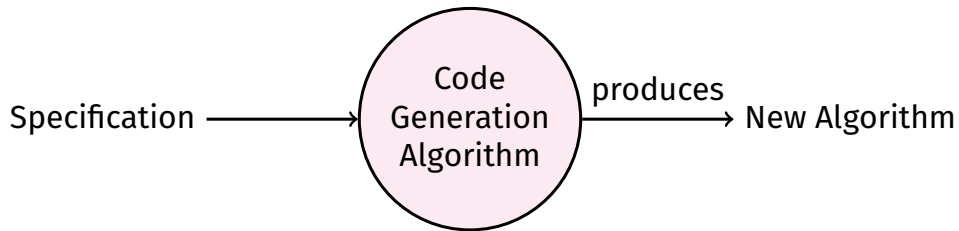
Research Contributions



Selected Publications

- 2026, 2 TOSEM (Q1) accepts with major revisions
- 2026, MSR (A)
- 2025, TACAS (A)
- 2025, AAI (top 20%) (A*)
- 2025, IST (Q1)
- 2022, AAI (top 20%) (A*)

PhD Motivation



PhD Motivation: In Practice

Specifications

Logic

$\forall a, b$
 $f(a, b) \geq a$
 $f(a, b) \geq b$
 $f(a, b) \in \{a, b\}$

Examples

$f(1, 5) = 5$
 $f(2, 1) = 2$
 $f(-3, -9) = -3$

Natural Language

‘Write a function that takes the maximum of its two arguments.’

Produced Algorithms

```
def max(a: int, b: int) ->int:
    if a <=b:
        return b
    else:
        return a
```

Program Synthesis: Problem

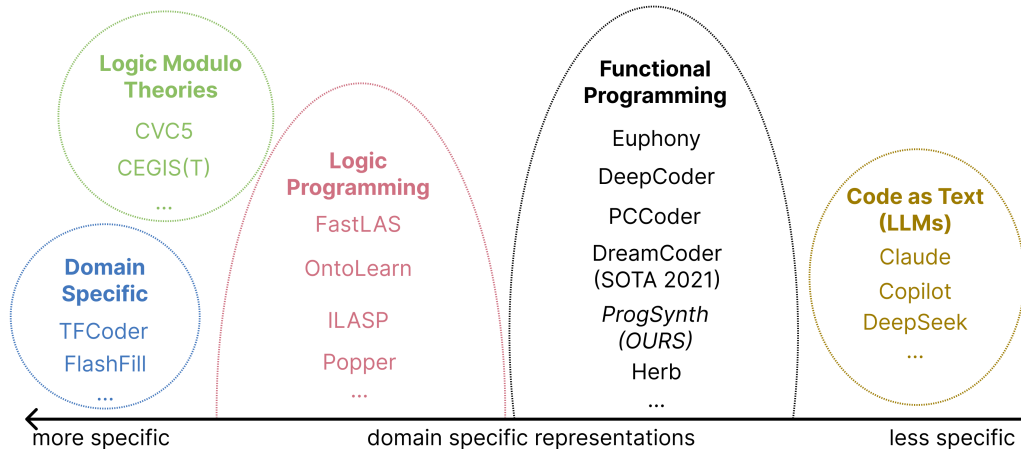
Input

- the search space G :
a deterministic tree grammar
- a specification $\mathcal{C}$:
checks if a program $p \in \mathcal{L}(G)$ matches the specification

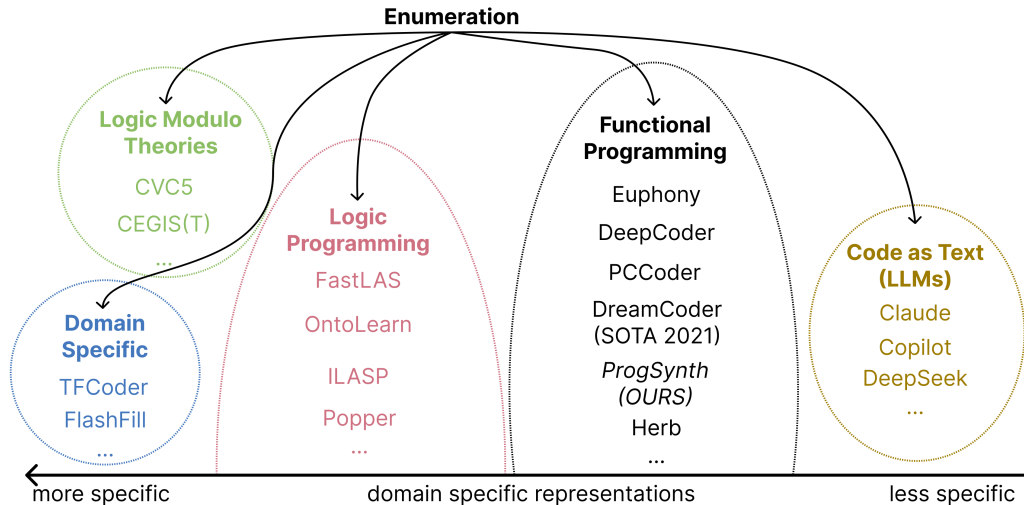
Output

- a program in the search space that matches the specification:
 $p \in \mathcal{L}(G)$ such that $\mathcal{C}(p) = \checkmark$

Program Synthesis: Frameworks



Program Synthesis: Frameworks



Program Synthesis: Enumeration Problem

Input

- the search space G :
a deterministic tree grammar with a cost for each tree
- a specification $\mathcal{C}$:
checks if a program $p \in \mathcal{L}(G)$ matches the specification

Goal

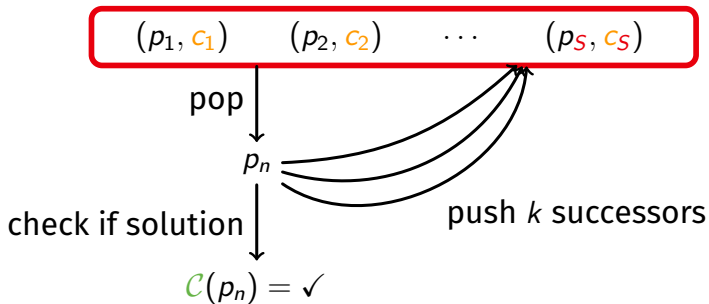
- enumeration of programs in order of non-decreasing costs

Delay

- time complexity between enumeration of two successive programs in terms of number of enumerated programs

Enumeration Algorithm Before Our Contribution

priority queue: S pairs of (program, cost)



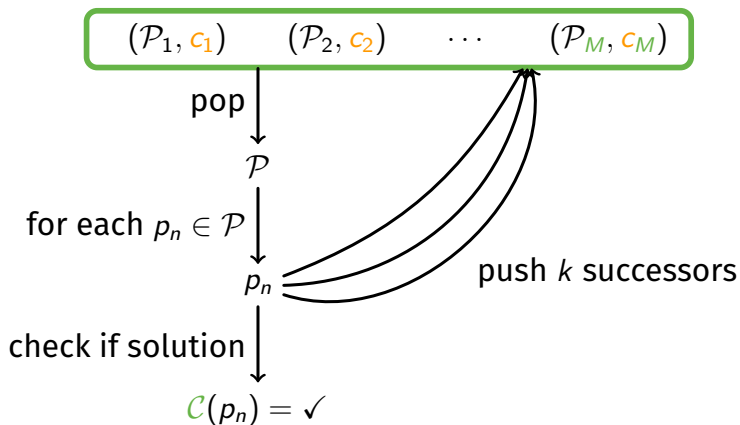
At step n

$$S = O(n)$$

$$k = O(1)$$

Our Contribution: Constant Delay

bucket queue: M buckets of (programs, cost)



At step n

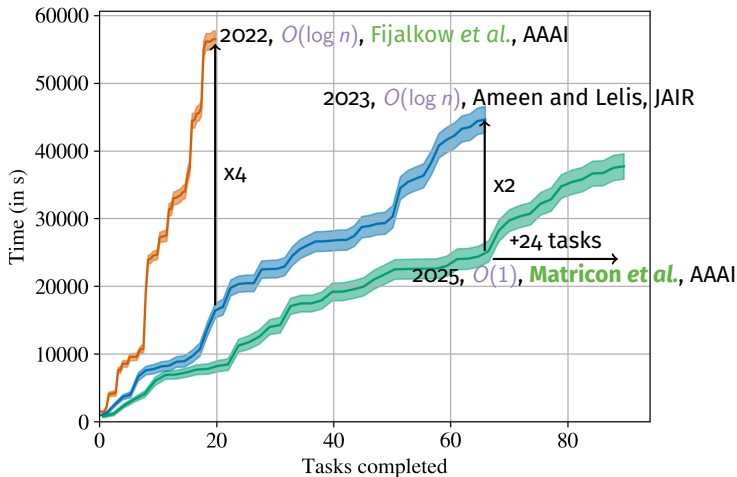
$$M = O(1)$$

$$k = O(1)$$

Our Contribution: Empirical Results

Major Papers

- 2017, ML + **Enum.**, Balog *et al.*, ICLR
- 2018, $O(\log n)$, Lee *et al.*, PLDI



Our Contribution: Summary

We prove bounded differences in cost:

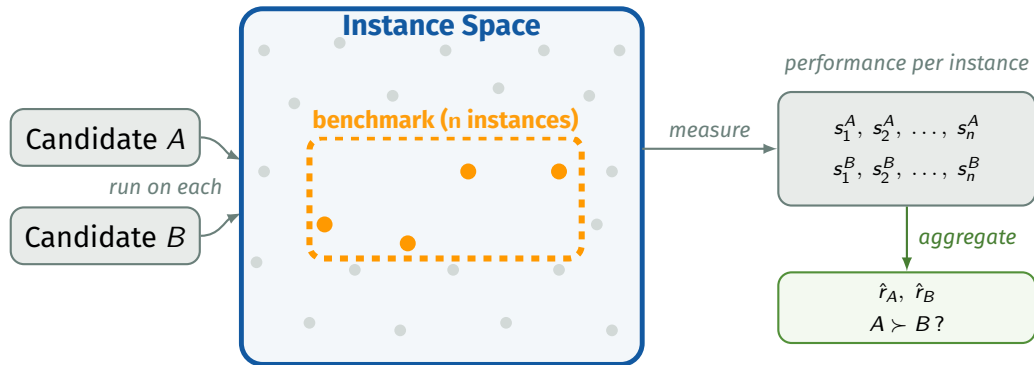
$$\exists M, \forall n, \text{cost_next}(p_n) - \text{cost}(p_n) \leq M$$

This implies: **priority queues** $O(\log n)$ $\rightarrow$ **bucket queues** $O(1)$

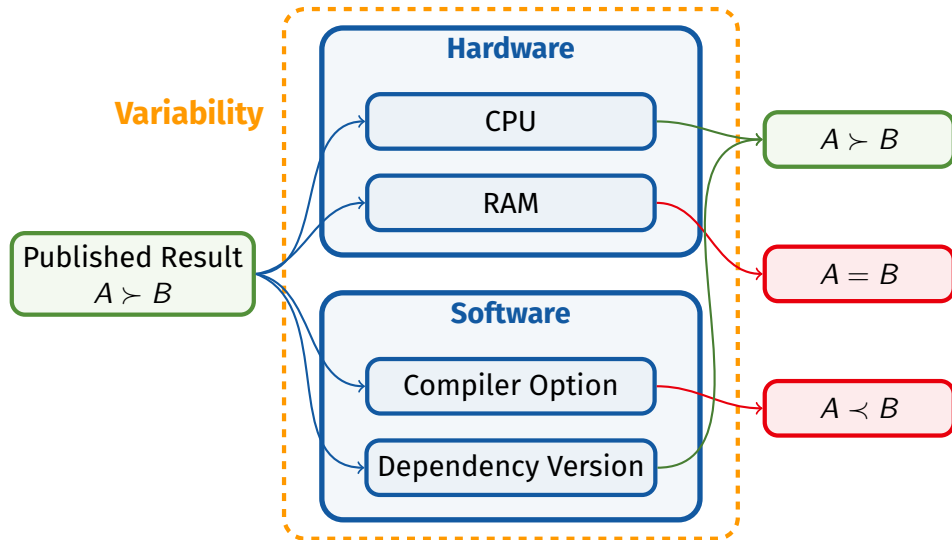
Impact

Published in **AAAI 2025** (+oral: 20% of accepted papers).
Fastest ranked enumeration for program synthesis **in practice**.
First algorithm with $O(1)$ delay $\rightarrow$ closes open question.

Evaluations



Evaluations: In Practice



My Research Project

Observations

- Exponential growth in scientific publications
- Reproducibility crisis in computational science
- Evaluations are a key aspect of reproducibility

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- **Exponential growth** in scientific publications
- **Reproducibility** crisis in computational science
- **Evaluations** are a key aspect of **reproducibility**

How to design **reliable and **efficient evaluations**?**

Reliable → formal guarantees

Efficient → reduce cost of **evaluations**

My Research Project

Observations

- **Exponential growth** in scientific publications
- **Reproducibility** crisis in computational science
- **Evaluations** are a key aspect of **reproducibility**

How to design **reliable** and **efficient evaluations**?

Reliable → formal guarantees

Efficient → reduce cost of **evaluations**

Main Challenges

SE Modelling the huge hardware and software **variability**

AI Predicting heterogeneous **evaluation** data

FM Giving formal guarantees while still being **efficient**

My Research Project

Current — no formal guarantees · costly · hard to reproduce

Candidates
+ Benchmark

Run **everything**

$\hat{r}$

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My Vision — reliable · efficient · reproducible

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Axis 1: Reliable
Bayesian U.Q.

select

observe

guarantee

Axis 2: Efficient
Adaptive eval.

stop?

Evaluate
(subset)

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Evaluate
(subset)

$$\mathbb{P}(r \in [\hat{r} - \Delta; \hat{r} + \Delta]) \geq 1 - \alpha$$

Classic Approach vs. Our Approach

Classic Approach

What is evaluated

Run all n instances — exhaustively

Our Approach

What is evaluated

Adaptive subset + early stopping
(~50% fewer runs)

Classic Approach vs. Our Approach

Classic Approach

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Statistical framework

assumes i.i.d. & normality

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Single environment assumed
results may not transfer

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Multi-environment hierarchical model
quantifies generalizability

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$$\hat{r} \pm z_{1-\alpha/2} \cdot \frac{s}{\sqrt{n}}$$

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Reproducibility

binary: reproduced or not

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What is evaluated

Adaptive subset + early stopping
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Statistical framework

Adjustable complexity

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Multi-environment hierarchical model
quantifies generalizability

Output

$$\mathbb{P}(r \in [\hat{r} - \Delta; \hat{r} + \Delta]) \geq 1 - \alpha$$

Reproducibility

graded: exact $\rightarrow$ ordinal $\rightarrow$ trend

Axis 1: Reliable Evaluations

Variability

CPU-A • GCC-12 • Linux

$\hat{r}_A = 0.82$ $\hat{r}_B = 0.75$ $A \succ B$

CPU-B • GCC-14 • Linux

$\hat{r}_A = 0.79$ $\hat{r}_B = 0.77$ $A \succ B$

CPU-C • Clang • macOS

$\hat{r}_A = 0.71$ $\hat{r}_B = 0.74$ $A \prec B$

inconsistent?

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Hierarchical Bayesian Model

Global

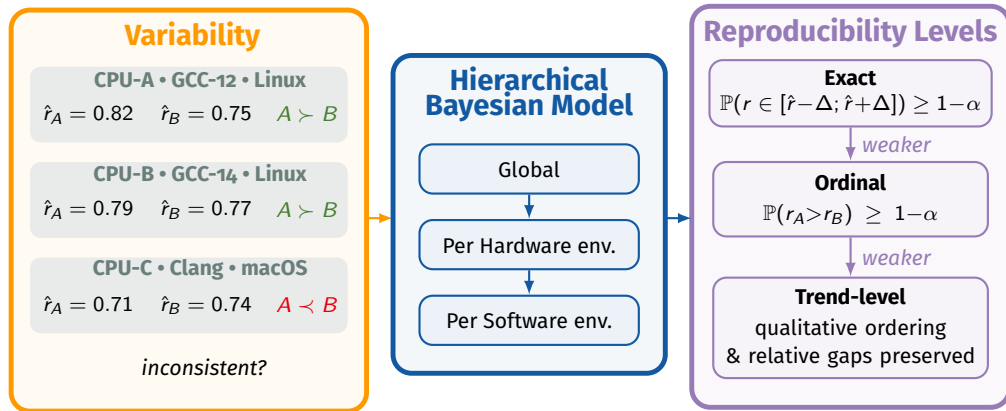


Per Hardware env.



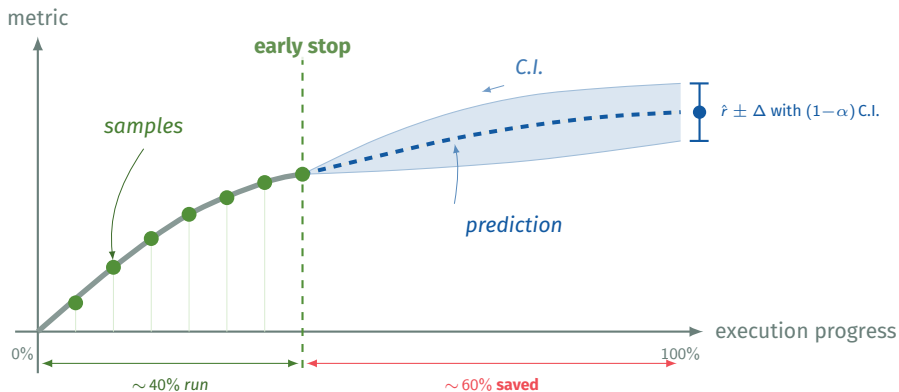
Per Software env.

Axis 1: Reliable Evaluations



Axis 2: Efficient Evaluations

Execution on an instance is not atomic



Integration

Transversal project with many applications in empirical fields

→ **CRIStAL**, UMR 9189, Lille, **Spirals** (SEEDS)

Frugality and Evaluations: Clément Quinton, Romain Rouvoy

Trust: Pierre Bourhis (Citadelle)

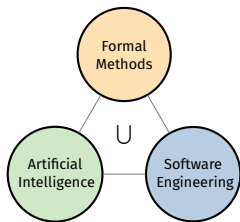
Statistics: Scool team

→ **IRISA**, UMR 6074, Rennes, **DiverSE** (rRaise)

Reproducibility: Mathieu Acher, Olivier Barais, Djamel E. Khelladi

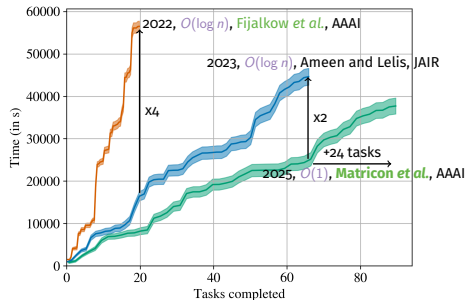
Frugality: *Romain Lefeuvre, Quentin Perez*

Théo Matricon



Selected Publications

- AI: 2 A^* →
- SE: 1+2 Q1, 1 A
- FM: 2 A



Research Project

